Lab work 2: The support vector machine

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We will consider the hard classification problem first, i.e., we suppose that all samples can be correctly classified using a linear separating hyperplane. In other words, we may consider a hyperplane parameterised by a vector $\boldsymbol{w} \in \mathbb{R}^n$ and a scalar $b \in \mathbb{R}$, defined by

$$\mathscr{H}_{\boldsymbol{w},b} \overset{\text{def.}}{=} \left\{ \boldsymbol{x} \in \mathbb{R}^n \middle| \boldsymbol{w}^t \boldsymbol{x} - b = 0 \right\} \ .$$

If we have a collection of labelled data points $\left\{ (\boldsymbol{x}_i, y_i) \in \mathbb{R}^n \times \{-1, 1\} \right\}_{i=1}^m$, the hyperplane $\mathscr{H}_{\boldsymbol{w},b}$ is separating the two classes labelled by -1 and $+1$ if and only if there exists a $\sigma \in \{-1, 1\}$ for which

$$\forall i, \sigma y_i \left(\boldsymbol{w}^t \boldsymbol{x}_i - b \right) > 0 \ .$$

The distance of a point $\boldsymbol{x} \in \mathbb{R}^n$ to the plane $\mathscr{H}_{\boldsymbol{w},b}$ is given by

$$\delta \left(\boldsymbol{x}, \mathscr{H}_{\boldsymbol{w},b} \right) \overset{\text{def.}}{=} \min_{\boldsymbol{z} \in \mathscr{H}_{\boldsymbol{w},b}} \| \boldsymbol{x} - \boldsymbol{z} \|$$

and the minimiser $\boldsymbol{z}^*$ of this problem also is a minimiser of

$$\frac{1}{2} \delta^2 \left(\boldsymbol{x}, \mathscr{H}_{\boldsymbol{w},b} \right) = \min_{\boldsymbol{z} \in \mathscr{H}_{\boldsymbol{w},b}} \frac{1}{2} \| \boldsymbol{x} - \boldsymbol{z} \|^2 \tag{1}$$

since $1/2\,(\cdot)^2$ is a strictly increasing function on $\mathbb{R}_+$. The latter is easier to manipulate, though.

1. Solve the above problem (1) using a Lagrangian. Can we state that solving the dual problem here also solves the primal problem? Why (not)?

2. Show that the following problem

$$\boldsymbol{w}^*, b^* \in \arg\max_{\boldsymbol{w},b} \min_{\{\boldsymbol{x}_i\}} \frac{1}{2} \delta^2 \left(\boldsymbol{x}_i, \mathscr{H}_{\boldsymbol{w},b} \right) \quad \text{subject to} \quad \forall i, y_i \left(\boldsymbol{w}^t \boldsymbol{x}_i - b \right) \geq 1$$

   leads to the hard–threshold (perfectly separable) support vector machine problem:

$$\boldsymbol{w}^*, b^* \in \arg\min_{\boldsymbol{w},b} \frac{1}{2} \| \boldsymbol{w} \|^2 \quad \text{subject to} \quad \forall i, y_i \left(\boldsymbol{w}^t \boldsymbol{x}_i - b \right) \geq 1 \ . \tag{2}$$

1 Solving the hard–threshold SVM problem

To solve the above problem (2), we use the Lagrangian

$$L(\boldsymbol{w}, b; \boldsymbol{\lambda}) = \frac{1}{2} \| \boldsymbol{w} \|^2 + \boldsymbol{\lambda}^t \left(\mathbb{1}_m + b\boldsymbol{y} - \boldsymbol{D}_{\boldsymbol{y}} \boldsymbol{X}^t \boldsymbol{w} \right) \tag{3}$$

where we have noted $\boldsymbol{D}_{\boldsymbol{y}} \in \mathbb{R}^{m \times m}$ for the diagonal matrix that contains $[y]_i$ on its diagonal positions $\left[\boldsymbol{D}_{\boldsymbol{y}} \right]_{i,i}$, $\boldsymbol{X} \overset{\text{def.}}{=} (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m) \in \mathbb{R}^{n \times m}$, and $\mathbb{1}_m \in \mathbb{R}^m$ is a vector of all ones.

3. **Primal stability conditions** and **complementary slackness**. Determine the stability conditions of (3) (vanishing gradients of L with respect to $\boldsymbol{w}$ and b) for the primal variables $\boldsymbol{w}^*$ as a function of the dual variable $\boldsymbol{\lambda}$. What is the expression of the dual function $g(\boldsymbol{\lambda}) = \min_{\boldsymbol{w},b} L(\boldsymbol{w}, b; \boldsymbol{\lambda})$?

4. **Primal feasibility**. How to find the optimal b^* as a function of $\boldsymbol{\lambda}^*$, $\boldsymbol{X}$, and $\boldsymbol{y}$?

5. **Dual feasibility**. Give the explicit expression for the dual optimisation problem. What constraints are imposed on $\boldsymbol{\lambda}$?

6. **A convex sub–problem**. When solving the dual problem, one is faced with the sum of a convex, differentiable function under an inequality constraint that acts on the canonical variables, hence the latter can easily be expressed with an indicator function. This gives rise to a typical forward–backward solver. Give the forward step with an appropriate step size.

7. **Implementation**. Solve the dual problem by a computer program (using a forward–backward algorithm with an appropriate step size for the forward step).
   Show the convergence by monitoring the cost as a function of the iteration number (log–log plot). To do so, generate two point clouds (points from the two classes which we will call + and −) which are separable (you may use two uniform distributions—or any other distributions with finite support—with different means $\mu_+ = (1/2, 0)$ and $\mu_- = (-1/2, 0)$) and test your code. To make sure your code works correctly, you may start testing on two points, one for each class for which the solution can be written in its analytic form: give the explicit solution.

The drawback of this method is that the classes must be perfectly separable, but in practice this is not always the case. This is where the soft–threshold comes into play.

2 Soft–threshold SVM problem

The soft–threshold problem considers that some of the vectors $\boldsymbol{x}_i$ may lie on the wrong side of the separating hyperplane or within the *margin*, in which case we have $y_i\left(\boldsymbol{w}^t\boldsymbol{x}_i - b\right) \leq 1$ for these points. If we want to minimise the number of vectors for which this is the case, we can use an ℓ_1-norm relaxation of this counting measure (the ℓ_0 pseudo-norm) and use this as a penalisation to our minimum norm objective:

$$\boldsymbol{w}^*, b^* = \arg\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2 + \lambda\frac{1}{m}\sum_{i=1}^{m}\max\left\{0, 1 - y_i\left(\boldsymbol{w}^t\boldsymbol{x}_i - b\right)\right\} \quad . \tag{4}$$

In order to deal with this problem, let us try a method to separate the variables in the penalty term. First, define the extended variables $\tilde{\boldsymbol{x}}_i \overset{\text{def.}}{=} \left(\boldsymbol{x}^t, -1\right)^t$, resulting in the augmented matrix $\tilde{X} \overset{\text{def.}}{=} (\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_m)$, and the extended vector of weights $\tilde{\boldsymbol{w}} \overset{\text{def.}}{=} \left(\boldsymbol{w}^t, b\right)^t$. Let $\boldsymbol{P}_{\boldsymbol{w}} \overset{\text{def.}}{=} \begin{pmatrix} \boldsymbol{I}_n & \boldsymbol{0}_n \\ \boldsymbol{0}_n^t & 0 \end{pmatrix}$ which is an idempotent operator, i.e., $\boldsymbol{P}_{\boldsymbol{w}}\boldsymbol{P}_{\boldsymbol{w}} = \boldsymbol{P}_{\boldsymbol{w}}$, where $\boldsymbol{I}_n$ is the identity matrix on $\mathbb{R}^n$ and $\boldsymbol{0}_n \in \mathbb{R}^n$ an all zero vector. Our problem (4) can now be rewritten as

$$\tilde{\boldsymbol{w}}^* = \arg\min_{\tilde{\boldsymbol{w}}} \frac{1}{2}\tilde{\boldsymbol{w}}^t\boldsymbol{P}_{\boldsymbol{w}}\tilde{\boldsymbol{w}} + \lambda\frac{1}{m}\mathbb{1}_m^t\max\left\{\boldsymbol{0}_m, \mathbb{1}_m - \boldsymbol{D}_{\boldsymbol{y}}\tilde{\boldsymbol{X}}^t\tilde{\boldsymbol{w}}\right\}$$

Now introduce $\boldsymbol{\zeta} \overset{\text{def.}}{=} \boldsymbol{D}_{\boldsymbol{y}}\tilde{\boldsymbol{X}}^t\tilde{\boldsymbol{w}}$. If $m \geq n+1$ and $\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^t$ is invertible, we may write $\tilde{\boldsymbol{X}}^{\#} \overset{\text{def.}}{=} \left(\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^t\right)^{-1}\tilde{\boldsymbol{X}}$ and we have $\tilde{\boldsymbol{X}}^{\#}\boldsymbol{D}_{\boldsymbol{y}}\boldsymbol{\zeta} = \tilde{\boldsymbol{X}}^{\#}\tilde{\boldsymbol{X}}^t\tilde{\boldsymbol{w}} = \tilde{\boldsymbol{w}}$.

8. Can you see why it is not necessary to consider the case $m \leq n$ here?

Now, with $\boldsymbol{X}^{\#} \overset{\text{def.}}{=} \left(\boldsymbol{X}\boldsymbol{X}^t\right)^{-1}\boldsymbol{X}$ and $\boldsymbol{\zeta} \overset{\text{def.}}{=} \boldsymbol{D}_{\boldsymbol{y}}\boldsymbol{X}^t\boldsymbol{w}$, we may rewrite problem (4) as

$$\boldsymbol{\zeta}^* = \arg\min_{\boldsymbol{\zeta}} \frac{1}{2}\left\|\boldsymbol{P}_{\boldsymbol{w}}\left(\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^t\right)^{-1}\tilde{\boldsymbol{X}}\boldsymbol{D}_{\boldsymbol{y}}\boldsymbol{\zeta}\right\|^2 + \lambda\frac{1}{m}\mathbb{1}_m^t\max\{\boldsymbol{0}_m, \mathbb{1}_m - \boldsymbol{\zeta}\}$$

9. With the slack variable $\boldsymbol{\zeta}$, we again have a minimisation problem that can be solved using a forward–backward algorithm. For the backward (proximal–point) step, one must compute the proximal operator of the function $f_1(z) = \max\{0, 1-z\}$, since the penalisation term is separable (each of the variables can be considered having an independent contribution). First show that f_1 is a convex function. Then give an explicit expression for $\text{prox}_{\frac{\lambda}{m}f_1}(z)$.

10. Implement the forward–backward algorithm with respect to $\boldsymbol{\zeta}$ choosing an appropriate step–size that guarantees a descent. Show the convergence by monitoring the cost as a function of the iteration number (log–log plot).
    Apply it to some toy data like a mixture of two–dimensional gaussians with different mean vectors for the two classes and sufficiently high variance such that overlap of the classes is almost unavoidable.

11. Using the instructions in the course document, load the MNIST dataset and apply the SVM method to two different classes of digits. Use a two-dimensional representation to show (a projection of) the optimal hyperplane.
    Optional: Perform a singular value decomposition of the matrix $\tilde{\boldsymbol{X}} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^t$ where the columns $\tilde{\boldsymbol{x}}_i$ of $\tilde{\boldsymbol{X}}$ contain the vectorised images of the training dataset (with $m > n$). Note that $\boldsymbol{U} = (\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n)$ with $\boldsymbol{u}_i \in \mathbb{R}^n$, $\boldsymbol{\Sigma} \in \mathbb{R}^{n\times n}$ is a diagonal matrix with positive entries, the diagonal being in decreasing order of magnitude, and

$\boldsymbol{V} = (\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n)$ with $\boldsymbol{v}_i \in \mathbb{R}^m$. The scaled two–dimensional representation of the vectorised image $\tilde{\boldsymbol{x}}_i$ is then given by $\left(\sigma_1^{-1}\boldsymbol{u}_1^t\tilde{\boldsymbol{x}}_i, \sigma_2^{-1}\boldsymbol{u}_2^t\tilde{\boldsymbol{x}}_i\right)^t$, hence for the matrix of all images, we find $\left(\sigma_1^{-1}\boldsymbol{u}_1, \sigma_2^{-1}\boldsymbol{u}_2\right)^t \tilde{\boldsymbol{X}}$ and for the vector of weights $(\sigma_1\boldsymbol{u}_1, \sigma_2\boldsymbol{u}_2)^t\,\tilde{\boldsymbol{w}}$ (which defines a straight line through the origin). You may want to compare learning the SVM in the two–dimensional reduced space or learning the SVM in the original, ambient space followed by a projection of the result into a two–dimensional space for visualisation and comparison (multi–dimensional scaling, PCA, ...).

3 Splitting the cost function with an augmented Lagrangian approach

When having introduced the slack variable $\boldsymbol{\zeta}$,we see that there is a problem when $\boldsymbol{X}\boldsymbol{X}^t$ is not full–rank. To overcome this, an alternative approach is to link the variables ζ and $\tilde{w}$ through a so–called augmented Lagrangian approach. First of all, note that we can rewrite our problem (4) as

$$\arg\min_{\tilde{\boldsymbol{w}},\boldsymbol{\zeta}} \frac{1}{2}\|\boldsymbol{P}_{\boldsymbol{w}}\tilde{\boldsymbol{w}}\|^2 + \lambda\frac{1}{m}\mathbb{1}_m^t \max\{0, 1-[\boldsymbol{\zeta}]_i\} \quad \text{subject to} \quad \boldsymbol{\zeta} = \boldsymbol{D}_{\boldsymbol{y}}\tilde{\boldsymbol{X}}^t\tilde{\boldsymbol{w}}$$

The Lagrangian of the problem is given by

$$L\left(\tilde{\boldsymbol{w}}, \boldsymbol{\zeta}; \boldsymbol{\eta}\right) = \frac{1}{2}\|\boldsymbol{P}_{\boldsymbol{w}}\tilde{\boldsymbol{w}}\|^2 + \lambda\frac{1}{m}\mathbb{1}_m^t \max\{0, 1-[\boldsymbol{\zeta}]_i\} + \boldsymbol{\eta}^t\left(\boldsymbol{\zeta} - \boldsymbol{D}_{\boldsymbol{y}}\tilde{\boldsymbol{X}}^t\tilde{\boldsymbol{w}}\right)$$

and add a penalisation term for some $\rho > 0$

$$L_\rho\left(\tilde{\boldsymbol{w}}, \boldsymbol{\zeta}; \boldsymbol{\eta}\right) = L\left(\tilde{\boldsymbol{w}}, \boldsymbol{\zeta}; \boldsymbol{\eta}\right) + \frac{\rho}{2}\left\|\boldsymbol{\zeta} - \boldsymbol{D}_{\boldsymbol{y}}\tilde{\boldsymbol{X}}^t\tilde{\boldsymbol{w}}\right\|^2 \quad .$$

One can now iteratively and alternatingly minimise with respect to $\tilde{\boldsymbol{w}}$ (gradient descent), then with respect to $\boldsymbol{\zeta}$ (proximal point), and finally perform a dual ascent on $\boldsymbol{\eta}$ by choosing the appropriate step size $\frac{1}{\rho}$ which will guarantee dual feasibility.

12. Implement the augmented Lagrangian approach to solve the soft–threshold SVM problem.